MIL-C-60167 (AR)
19 October 1964
SUPERSEDING
FA-PD No. SP-1960
10 February 1955

MILITARY SPECIFICATION
CARTRIDGE, CALIBER .50, TEST, HIGH PRESSURE, T251

1. SCOPE

1.1 This specification covers a cartridge for proof testing Caliber .50 Spotter-Tracer weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-636 - Visual Inspection Standards for Small Arms Ammunition through Caliber .50

MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

D10534770 - Cartridge, Caliber .50, Test, High Pressure, T251

F10534778 - Packing and Marking, Cartridges, Caliber .50, Test, High Pressure, T251; Cartons; Box, Ammunition, M2A1; Box, Wirebound

IEL-10534770- Inspection Equipment List for Cartridges, Caliber .50, Test, High Pressure, T251

FSC 1305

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PUBLICATIONS

Munitions Command
AMCR 700-700, Vol. III - Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with Drawing D10534770, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Bullet extraction.- The force required to extract the bullet from the cartridge case shall be not less than 200 pounds.

3.4 Residual stress.- The cartridge case shall not split when subjected to one per cent mercurous nitrate solution for fifteen minutes.

3.5 Chamber pressure.- The average chamber pressure of the sample cartridges, conditioned at $70^{\circ} \pm 2^{\circ}$ Fahrenheit (F), shall be 53,000 pounds per square inch (psi) plus or minus 2,000 psi.

3.6 Functional casualty.- The cartridge shall function without casualty.

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specification and the following:

3.7.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.7.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 Initial production sample.- Prior to the initiation of regular production, the supplier shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 300 cartridges. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 per cent and the AQL for the minor class shall be 1.50 per cent.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

Table I

No.	Defect and Method of Inspection	Major	Minor
	Visual 1/		
	Cartridge		
1	Discolored, dirty, oily, smeared		X
2	Corroded or stained, if etched		X

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Table I (Cont'd)

No.	Defect and Method of Inspection	Major	Minor
	Visual 1/		
	Case		
4	Round head		X
5	Dent		X
6	Split case 2/	X	
7	Perforated case	X	
8	Draw scratch		X
10	Beveled underside of head		X
11	Case mouth not crimped in cannelure		X
12	Scaly metal		X
13	No chamfer on head (rim)		X
14	Fold		X
15	Wrinkle		X
16	Buckle		X
17	Bulge		X
18	Illegible or missing head stamp		X
19	Defective head		X
20	Defective mouth		X
21	No visible evidence of mouth anneal	X	
21A	Coating missing	X	
	Bullet		
24	Split bullet jacket		X
25	Loose bullet	X	
26	Missing cannelure		X
27	Scaly metal		X
29	Exposed steel (clad jacket)		X
31	Defective cannelure		X
	Primer		
32	No primer	X	
33	Cocked primer	X	
34	Inverted primer	X	
35	Loose primer	X	
36	Nicked or dented primer		X
37	No waterproofing material (primer pocket joint)		X
38	Defective crimp		X
	Gaging		
39	Total length	X	
40	Cartridge profile failure (requiring more than 80 lbs. dead weight to insert in profile and alignment gage)	X	

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Table I (Cont'd)

No.	Defect and Method of Inspection	Major	Minor
	Gaging		
41	Diameter of extractor groove		X
43	Diameter of head		X
44	Thickness of head		X
45	Length to shoulder datum	X	
46	Depth of primer	X	
	Weighing		
47	Weight, min.	X	

1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 38 with the exception of defect No. 21A. In the event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

2/ A split in any location and with or without evidence of propellant loss shall be classed as a major defect.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

Table II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.3
Residual stress (Mercurous nitrate) 1/	50	3.4
Chamber pressure 2/	25	3.5
Function and casualty 3/		3.6

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

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2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

3/ Observation for function and casualty shall be made during firing of the chamber pressure test. The lot shall be rejected when firing defects exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample consisting of double the quantity of cartridges specified for the chamber pressure test shall be fired. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

Table III

Defects	Acceptance Number	
	First Sample	Cumulative (1st & 2nd Sample)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Blown primer or primer falls out of pocket on retraction of bolt	0	1
4. Case casualties		
a. Split (longitudinal)	1	2
b. Complete rupture (circumferential)	0	1
c. Partial rupture (circumferential)	1	2

1/ No second sample permitted. Lot shall be rejected.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of any type other than HPT shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing, and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

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4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-10534770.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Residual stress (Mercurous nitrate).- The test shall be conducted in accordance with AMCR 700-700 Vol. III.

4.4.3 Chamber pressure.- The test shall be conducted in accordance with AMCR 700-700 Vol. III.

4.4.4 Function and casualty.- Observation for firing defects listed in Table III shall be made during the firing of the chamber pressure test. When testing of a second sample is required, the test shall be conducted in accordance with the procedure prescribed for the chamber pressure test in AMCR 700-700, Vol. III, except that all references pertaining to obtaining of chamber pressure data shall be disregarded.

4.4.5 Defect penalty.- If, in the chamber pressure test, a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. The chamber pressure test shall not be penalized, but the acceptance or initial production sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.- (Worldwide shipment). The cartridges shall be packed in accordance with Drawing F10534778.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids or request for proposal and contracts or orders will specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

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6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

Custodian:

Army - AR

Preparing Activity:

Army - AR

Project No. 1305-A317

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-60167	CARTRIDGE, CALIBER .50, TEST, HIGH PRESSURE, T251

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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